

Summary of Revisions

Addressing Concerns by Reviewer TjL

How to read this

Dear Reviewer TjL, we thank you again for your thoughtful feedback. We cannot upload a full revised manuscript within the discussion window, so we summarize the changes prompted by the review. Each *Before* block reproduces the current submission verbatim; each *After* block shows the revised text, with [new or revised wording in blue](#).

Correspondence to the review.

- **Formal model presented as validated (§2.2–2.3):** Revisions 1 and 2.
- **Related work and contribution framing (W1):** Revisions 3 and 4.
- **Tokenizer averaging in Tables 4–5 (W3):** Revision 5.
- **Missing sampling baselines (W4):** Revision 6.
- **Class balance in classifier training (question):** Revision 7.

1 Recast §2.2 as motivation, not a validated model

Addresses: The concern that the paper presents a formal model of human token selection and claims its results support it.

Location: Section 2.2, the two paragraphs beginning “Human language production can be conceptualized...”.

Before — current submission (verbatim)

Human language production can be conceptualized as drawing from a distribution $P_{\mathcal{H}}(x_t \mid \mathbf{x}_{<t}, \mathcal{I})$, where $\mathcal{I}$ captures latent parameters encoding the speaker’s communicative intent, including discourse goals, audience modeling, register constraints, and pragmatic considerations (Levelt, 1989; Sperber & Wilson, 1986). While neither $\mathcal{I}$ nor the functional form of $P_{\mathcal{H}}$ are directly observable, we can characterize the human selection mechanism conceptually through its realizations.

A human writer typically chooses words based on qualitative criteria such as contextual appropriateness, grammatical well-formedness, and stylistic expectations (Pickering & Garrod, 2007; Dell, 1986). Unlike the likelihood-driven selection of LLMs, human selection is not primarily governed by statistical frequency. Consequently, these two mechanisms produce distinct partitions of the vocabulary for any given context; the human truncation set,

$\mathcal{T}_{\mathcal{H}}(\mathbf{x}_{<t})$, does not necessarily correspond to the model’s truncation set, $\mathcal{T}_{\theta}(\mathbf{x}_{<t})$.

After — revised

Human language production can be conceptualized as drawing from a distribution $P_{\mathcal{H}}(x_t \mid \mathbf{x}_{<t}, \mathcal{I})$, where $\mathcal{I}$ captures latent parameters encoding the speaker’s communicative intent, including discourse goals, audience modeling, register constraints, and pragmatic considerations (Levelt, 1989; Sperber & Wilson, 1986). **We invoke $P_{\mathcal{H}}$ as a conceptual reference rather than an estimand: it defines what makes a token available to a writer, and thereby delimits the blind spot we set out to measure.**

A human writer typically chooses words based on qualitative criteria such as contextual appropriateness, grammatical well-formedness, and stylistic expectations (Pickering & Garrod, 2007; Dell, 1986), **which need not align with a model’s probability ranking: a precise but rare word can be communicatively apt yet statistically unlikely. The two mechanisms therefore partition the vocabulary differently:** the human truncation set $\mathcal{T}_{\mathcal{H}}(\mathbf{x}_{<t})$ does not necessarily correspond to the model’s truncation set $\mathcal{T}_{\theta}(\mathbf{x}_{<t})$, **and it is this decoding-time mismatch that our analysis targets.**

Effect: $P_{\mathcal{H}}$ is now framed as a conceptual reference, and the empirical target is named as the decoding-time mismatch. The earlier “not primarily governed by statistical frequency” claim is replaced by the weaker, accurate statement that human choices need not align with the model’s ranking.

2 Simplify §2.3: remove the ergodicity/stationarity apparatus

Addresses: The concern that the §2.3 derivation is unnecessary and that stationarity/ergodicity are unjustified.

Location: Section 2.3, entire body.

Before — current submission (verbatim)

We are ultimately interested in quantifying the extent to which human-generated tokens fall into the blind spot at any generation step t . Formally, we seek to estimate the probability of the event that, given the context $\mathbf{x}_{<t}$, a random token X_t lies in the intersection of the human inclusion set and the model’s exclusion region under the true token distribution, i.e., the distribution of the human speaker:

$$\begin{aligned} P_{\mathcal{H}}(X_t \in \mathcal{T}_{\mathcal{H}}(\mathbf{x}_{<t}) \cap \mathcal{E}_{\theta}(\mathbf{x}_{<t})) \\ = P_{\mathcal{H}}(X_t \in \mathcal{T}_{\mathcal{H}}(\mathbf{x}_{<t}) \wedge X_t \notin \mathcal{T}_{\theta}(\mathbf{x}_{<t})). \end{aligned}$$

Let $\mathbf{x}^H = (x_1^H, \dots, x_T^H)$ denote an observed sequence of human-generated tokens. To bridge the gap between empirical observation and theoretical probability, we ground our analysis in information theory (Shannon, 1951; Cover & Thomas, 1999). Acknowledging that natural language is non-stationary over arbitrary timescales, we assume local stationarity within homogeneous thematic contexts, such as formal journalism, encyclopedic discourse, and scientific writing. Accordingly, we model the human token sequence as a realization of

a stochastic process $\{X_t^H\}_{t \in \mathbb{Z}}$ that admits an ergodic approximation, thereby justifying the substitution of time averages for ensemble averages over the finite sample of our corpus. We define the *exclusion indicator* Z_t as a binary random variable representing the event that a human token X_t^H falls outside the model’s truncation set given the history $\mathbf{X}_{<t}^H$:

$$Z_t = \mathbf{1} [X_t^H \notin \mathcal{T}_\theta(\mathbf{X}_{<t}^H)],$$

where $\mathbf{1}[\cdot]$ is the indicator function. Since the mapping from the sequence X^H to the exclusion status is a time-invariant transformation, the derived process $\{Z_t\}_{t \in \mathbb{Z}}$ inherits the properties of stationarity and ergodicity from $\{X_t^H\}$. Under the human distribution P_H , the event $X_t^H \in \mathcal{T}_H(\mathbf{X}_{<t}^H)$ occurs with probability 1 by definition.

Consequently, the joint probability of the blind spot simplifies to the marginal probability of exclusion:

$$\begin{aligned} P_H(X_t^H \in \mathcal{T}_H(\mathbf{X}_{<t}^H) \cap \mathcal{E}_\theta(\mathbf{X}_{<t}^H)) &= P_H(X_t^H \notin \mathcal{T}_\theta(\mathbf{X}_{<t}^H)) \\ &= P(Z_t = 1) \end{aligned}$$

By the ergodic theorem, as $T \rightarrow \infty$, the time average of the ergodic process $\{Z_t\}$ converges to its expected value $E[Z_t] = P(Z_t = 1)$. Therefore, for a sufficiently large corpus T , we can estimate this probability via the empirical *human token exclusion rate*:

$$\mathcal{R}_{\text{excl}}(\mathbf{x}^H; \theta) = \frac{1}{T} \sum_{t=1}^T \mathbf{1} [x_t^H \notin \mathcal{T}_\theta(\mathbf{x}_{<t}^H)]$$

This rate serves as a tractable proxy for the volume of the blind spot, measuring the proportion of communicatively valid tokens that are rendered unreachable under a specific neural text generator θ .

After — revised

We quantify how often human-selected tokens fall into the blind spot. Because $\mathcal{T}_H$ is, by construction, the set of tokens a writer selects, every observed human token lies in $\mathcal{T}_H$, and the blind-spot event coincides with exclusion by the model:

$$x_t^H \in \mathcal{T}_H(\mathbf{x}_{<t}^H) \cap \mathcal{E}_\theta(\mathbf{x}_{<t}^H) \iff x_t^H \notin \mathcal{T}_\theta(\mathbf{x}_{<t}^H).$$

Its magnitude is the rate at which human tokens fall outside the model’s truncation set,

$$\mathcal{R}_{\text{excl}}(\mathbf{x}^H; \theta) = \frac{1}{T} \sum_{t=1}^T \mathbf{1} [x_t^H \notin \mathcal{T}_\theta(\mathbf{x}_{<t}^H)],$$

a direct sample statistic that counts how often a writer’s token would have been unreachable under generator θ , and requires no assumptions about the process that produced the text.

Effect: The stochastic-process, stationarity, and ergodic-theorem steps are removed; the exclusion rate now follows in one line from the definition of $\mathcal{T}_H$. The estimand and all downstream results are unchanged.

3 Add a “Positioning” paragraph repositioning the contribution

Addresses: W1: framing as empirical characterization rather than discovery, with the prior works cited.

Location: Section 1, inserted after the paragraph ending “...neither architectural advances nor alignment procedures eliminate the underlying mismatch.” Nothing is removed.

Before — current submission (verbatim)

(New paragraph; no existing text is removed.)

After — revised

Positioning. That likelihood-based decoding can exclude tokens humans use, and that this bears on detectability, is not itself new: it surfaces in the quality–detectability trade-off of Ippolito et al. (2020), in the truncation analyses of Hewitt et al. (2022) and Finlayson et al. (2024), and in the concurrent study of Dubois et al. (2025). Our contribution is to characterize the effect empirically and at scale: (i) a unified measurement across eight open, proprietary, and non-Transformer systems, five decoding strategies, and 53 configurations; (ii) a part-of-speech and domain-level decomposition of the excluded tokens (Section 5.3); (iii) a within-family analysis of how truncation-set overlap changes with scale (Appendix A.13); and (iv) a controlled separation of the decoding mechanism from model scale, architecture, and alignment as drivers of detectability (Section 5.4).

4 Add a related-work paragraph positioning the prior work

Addresses: W1: the missing references and their relation to our work.

Location: Section 3, inserted after the paragraph ending “...also introduce systematic deviations from human text.”

After — revised

Several studies bear directly on our analysis. Ippolito et al. (2020) show that decoding choices tuned to fool human readers introduce statistical regularities that make automatic detection easier; we account for that trade-off mechanistically, tracing it to the tokens truncation removes. Hewitt et al. (2022) cast truncation as desmoothing, recovering the support of the true distribution, and propose η - and ϵ -sampling to do so more judiciously; we take the complementary, diagnostic view, asking what such truncation excludes relative to human text, and adopt their methods as baselines (Section 4.2). Finlayson et al. (2024) trace degeneration to the softmax bottleneck and derive a principled truncation criterion; their treatment is largely theoretical, whereas we measure empirically what truncation removes and how it drives detectability. Closest to our setting, Dubois et al. (2025) show concurrently that small changes in sampling parameters can sharply alter detector accuracy; our results concur that decoding governs detectability, and add a mechanistic account in terms of token exclusion, a linguistic decomposition of the excluded set, and coverage of proprietary and non-Transformer models.

5 Clarify that exclusion is computed within each model’s tokenizer

Addresses: W3: exclusion rates in Tables 4–5 are averaged across models with different tokenizers.

(a) Appendix text. Location: Appendix A.2, opening paragraph.

Before — current submission (verbatim)

Tables 4 and 5 quantify the magnitude of the blind spot discussed in Section 5.1. Table 4 reports human token exclusion rates across truncation configurations, while Table 5 shows the distribution of human token ranks.

After — revised

Tables 4 and 5 quantify the magnitude of the blind spot discussed in Section 5.1. Table 4 reports human token exclusion rates across truncation configurations, while Table 5 shows the distribution of human token ranks. **Exclusion is computed entirely within each model’s own tokenizer: each text tokenized by that model, ranks taken under its distribution, exclusion judged against its truncation set, so ranks are never compared across vocabularies; we average only the resulting per-model rates, which are unitless.**

(b) Table 4 caption.

Before — current submission (verbatim)

Human token exclusion rates $\mathcal{R}_{\text{excl}}$ across truncation configurations. Values are averaged over five language models (GPT2-XL, Qwen2-7B, Mistral-7B, LLaMA3-8B, Falcon2-11B) and all texts in each dataset split; 95% confidence intervals are computed over this model $\times$ text distribution. Values indicate the percentage of tokens in human text that would fall in the blind spot under each setting.

After — revised

Human token exclusion rates $\mathcal{R}_{\text{excl}}$ across truncation configurations. Values are averaged over five language models (GPT2-XL, Qwen2-7B, Mistral-7B, LLaMA3-8B, Falcon2-11B) and all texts in each dataset split; 95% confidence intervals are computed over this model $\times$ text distribution. Values indicate the percentage of tokens in human text that would fall in the blind spot under each setting. **Rates are computed within each model’s own tokenizer; ranks are never compared across vocabularies, and only the resulting unitless rates are averaged.**

6 Add tail-recovering sampling baselines to the protocol

Addresses: W4: missing locally typical, η -, and ϵ -sampling baselines.

Location: Section 4.2, appended to the final sentence.

Before — current submission (verbatim)

Top- p sampling (Holtzman et al., 2020) samples from the nucleus, with $p \in \{0.6, 0.7, 0.8, 0.9, 0.95\}$, this yields 53 configurations per model and accounts for over 1.8 million generated texts, which we investigate along with their human-written counterparts.

After — revised

Top- p sampling (Holtzman et al., 2020) samples from the nucleus, with $p \in \{0.6, 0.7, 0.8, 0.9, 0.95\}$, this yields 53 configurations per model and accounts for over 1.8 million generated texts, which we investigate along with their human-written counterparts. We additionally evaluate locally typical sampling (Meister et al., 2023) and η - ϵ -sampling (Hewitt et al., 2022), which truncate by information content rather than rank or mass, comparing them against top- k and top- p at matched exclusion (Appendix A.7).

7 Address class balance in classifier training

Addresses: The question of whether the machine/human class imbalance drives separability.

Location: Section 4.5, appended to the final sentence.

Before — current submission (verbatim)

The use of simple classifiers is intentional: high classification performance with minimal features would indicate that machine text exhibits systematically different statistical properties.

After — revised

The use of simple classifiers is intentional: high classification performance with minimal features would indicate that machine text exhibits systematically different statistical properties. Because AUC-ROC is invariant to the class prior, the human/machine imbalance cannot inflate the reported separability; downsampling machine examples to the human count leaves AUC-ROC essentially unchanged.

New references added

The following are added (Revisions 3, 4, 6); abstracts omitted for brevity.

```
@inproceedings{dubois-etal-2025-sampling,
  title      = {How Sampling Affects the Detectability of Machine-written Texts:
               A Comprehensive Study},
  author     = {Dubois, Matthieu and Yvon, Fran\c{c}ois and Piantanida, Pablo},
  booktitle  = {Findings of the Association for Computational Linguistics: EMNLP 2025},
  pages      = {11369--11387},
  year       = {2025}
}
@inproceedings{ippolito2020automatic,
```

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    title      = {Automatic Detection of Generated Text is Easiest when Humans are Fooled},
    author     = {Ippolito, Daphne and Duckworth, Daniel and Callison-Burch, Chris and Eck, Doug},
    booktitle  = {Proceedings of the 58th Annual Meeting of the Association for
                  Computational Linguistics},
    pages      = {1808--1822},
    year       = {2020}
}
@inproceedings{hewitt2022truncation,
    title      = {Truncation Sampling as Language Model Desmoothing},
    author     = {Hewitt, John and Manning, Christopher D. and Liang, Percy},
    booktitle  = {Findings of the Association for Computational Linguistics: EMNLP 2022},
    pages      = {3414--3427},
    year       = {2022}
}
@inproceedings{finlayson2024closing,
    title      = {Closing the Curious Case of Neural Text Degeneration},
    author     = {Finlayson, Matthew and Hewitt, John and Koller, Alexander and
                  Swayamdipta, Swabha and Sabharwal, Ashish},
    booktitle  = {International Conference on Learning Representations},
    pages      = {12295--12313},
    year       = {2024}
}

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